

8. The **epicentre** is the point on the Earth's surface directly above an earthquake. Seismic stations detect earthquakes by the tracings made on seismographs.

(a) Surface, P and S waves are three types of earthquake waves.  
Tick (✓) the boxes next to the **three** correct statements about earthquake waves. [3]

Surface waves travel the fastest ☐

S waves travel on the surface of the Earth ☐

S waves are transverse waves ☐

P waves travel through solids and liquids ☐

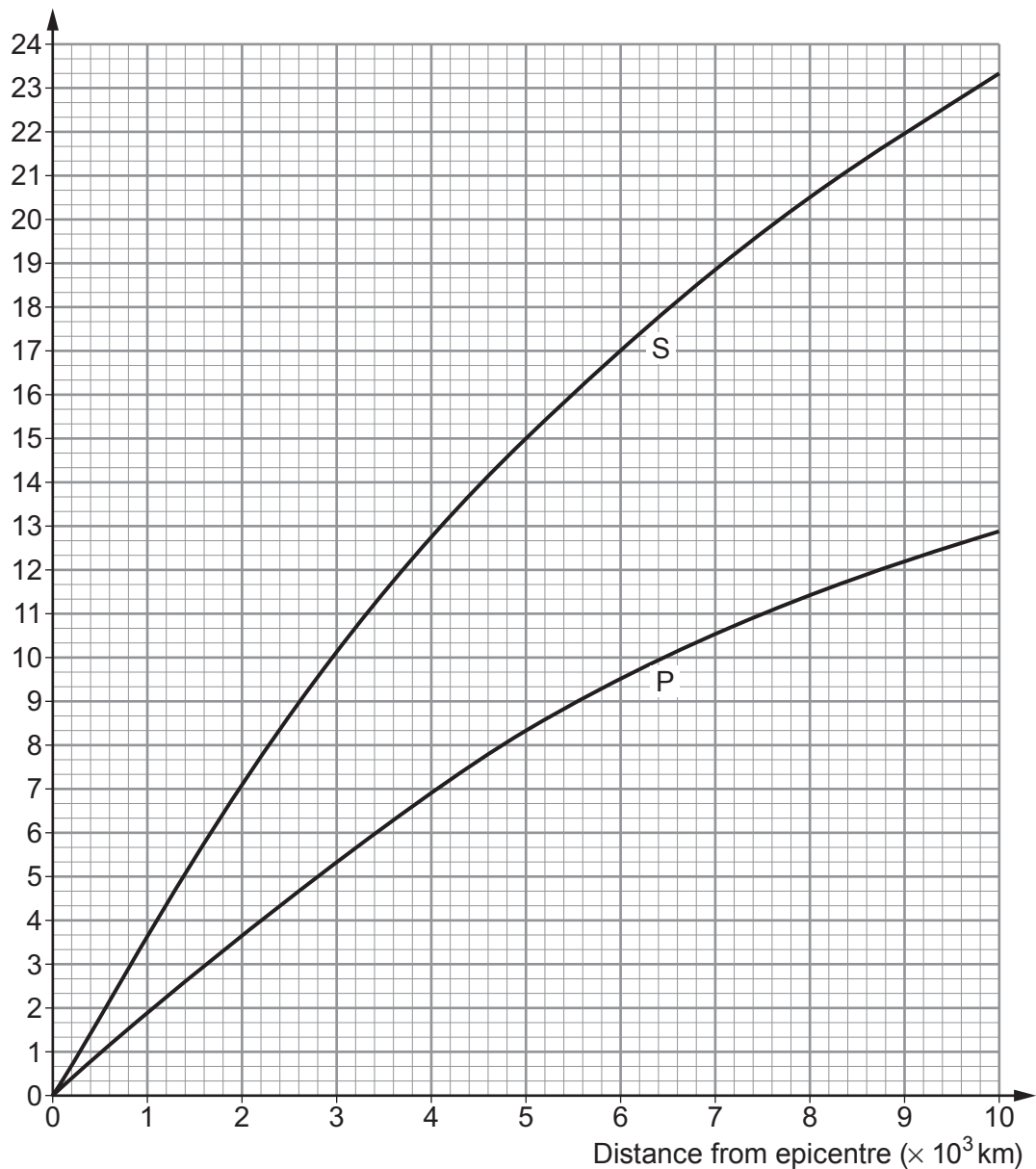
P waves are longitudinal waves ☐

S waves cause the most damage ☐



- (b) The graph shows the time taken by P and S waves to travel different distances from the epicentre.

Time (min)



Each small square on the time axis represents 20 s.

- (i) Use the graph to answer the following questions.

I. State the time it takes for a P wave to travel  $5 \times 10^3$  km from the epicentre.

[1]

time = ..... min ..... s

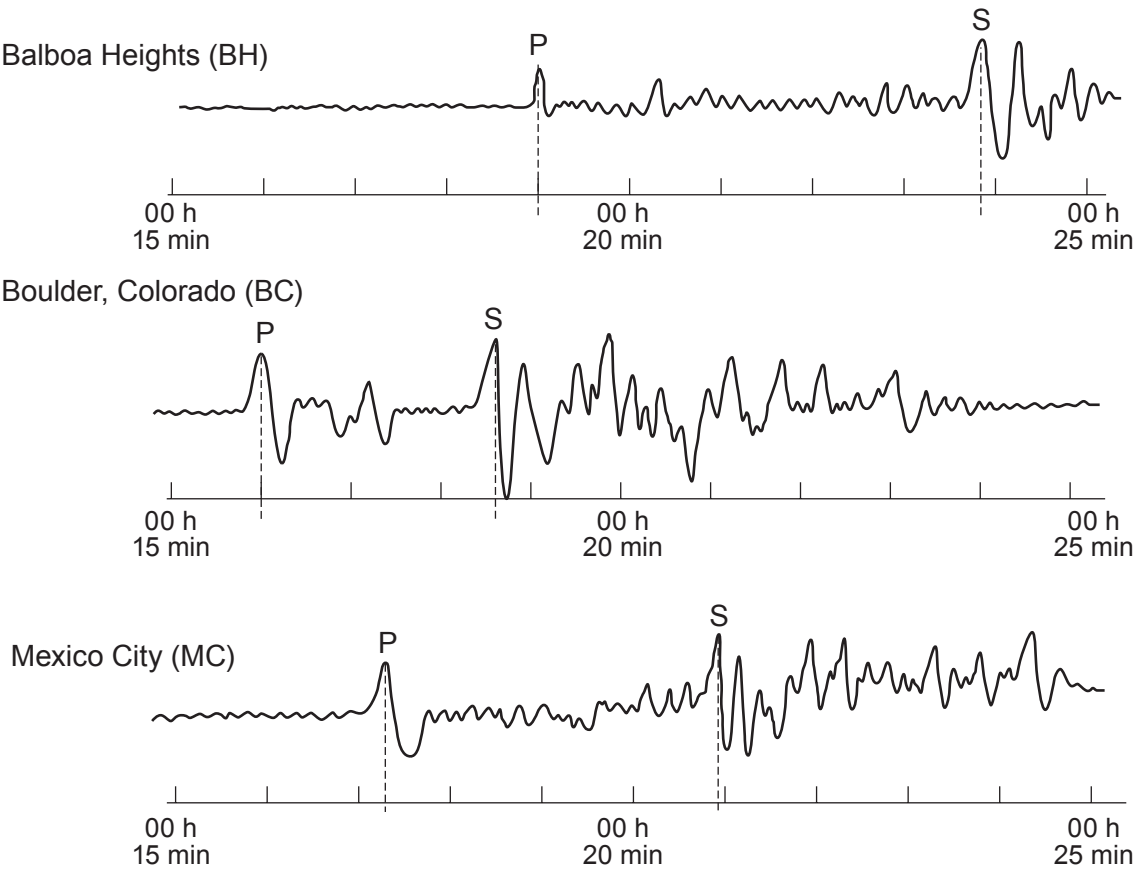
II. State the **extra** time it takes S waves to travel  $5 \times 10^3$  km from the epicentre.

[1]

time = ..... min ..... s



(ii) Study the three seismograph tracings below. Tracings made at three separate seismic stations are needed to locate an earthquake epicentre. P shows the arrival of P waves and S shows the arrival of S waves.



Use the information in the graph and tracings to **complete the table**. [3]

| City                   | Arrival time of P waves<br>(h:min:s) | Arrival time of S waves<br>(h:min:s) | Time difference for P and S waves<br>(h:min:s) | Distance to epicentre<br>( $\times 10^3$ km) |
|------------------------|--------------------------------------|--------------------------------------|------------------------------------------------|----------------------------------------------|
| Balboa Heights (BH)    | 00:19:00                             | 00:23:50                             | 00:04:50                                       | 3.2                                          |
| Boulder, Colorado (BC) | ..... : ..... : .....                | 00:18:40                             | ..... : ..... : .....                          | .....                                        |
| Mexico City (MC)       | 00:17:15                             | 00:20:55                             | 00:03:40                                       | 2.2                                          |

**TURN OVER FOR THE REST OF THE QUESTION**



- (iii) The P waves arriving at Balboa Heights (BH) took 6 **minutes** to travel from the epicentre. Use an equation from page 2 to calculate the speed of the P waves arriving at Balboa Heights in **km/s**. [Note that  $3.2 \times 10^3 \text{ km} = 3\,200 \text{ km}$ ] [2]

Speed = ..... km/s

- (iv) The data is used to locate the epicentre of the earthquake. **Indicate with crosses (X) on the diagram opposite two** possible positions for the location of the earthquake. [1]

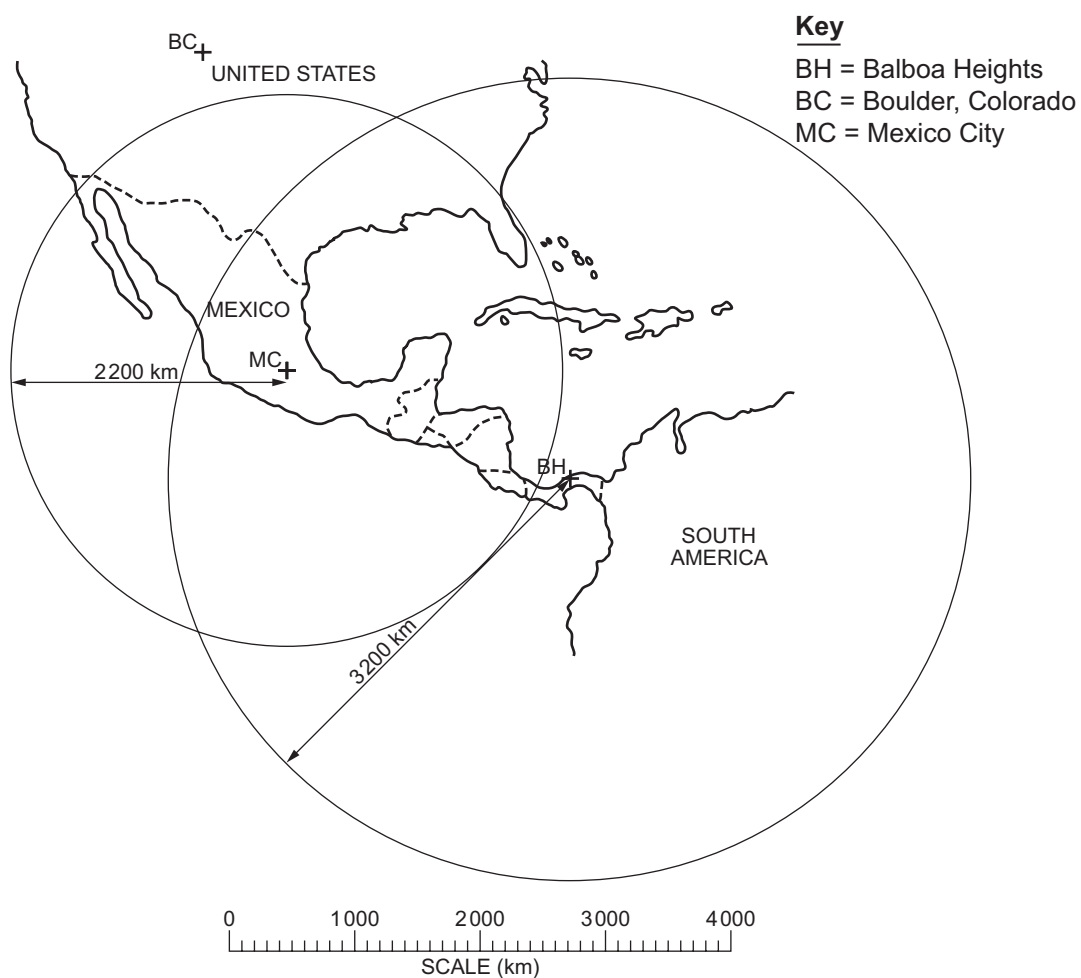
- (v) Use the data for Boulder Colorado (BC) to show clearly **on the diagram opposite** the actual location of the epicentre. Justify how you have arrived at your answer. [2]

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**END OF PAPER**

